



Moon Buggy

Lockheed Martin and GM have unveiled plans to make an autonomous vehicle for astronauts. <http://digit.in/jun21-55>



Rover captures clouds on Mars

NASA's Curiosity rover has captured images of shining, iridescent, noctilucent clouds on Mars. <http://digit.in/jun21-56>

massive planets that orbit very close to the host star. We have to note here that the transit method used in most of the large scale exoplanet surveys that have found these 150 or so Hot Jupiters are particularly successful at finding large planets close to their host stars. There are several theories on how hot Jupiters are formed, one being that they are formed close to the host stars which would mean nearby smaller planets could remain in orbit, or they could have fallen inwards after forming at a distance from the stars, which would mean that the planets on the way between where the gas giant formed and the orbit it occupies, would have been eaten up by the gas giant. Another possibility of an elliptical inward journey of a giant planet is that other planets in the way get ejected from the system. Scientists have also found wandering, lonely planets, far away from any systems.

The finding of such aberrations meant that researchers had to go back to the drawing board to explain the whole process of star system formation to better explain all the planets and stars that we know about. While making guesses from observations of mature systems is the main approach, a new one involves studying the protoplanetary discs to catch the universe in the act of birthing star systems. Astronomers have identified the nearby protoplan-

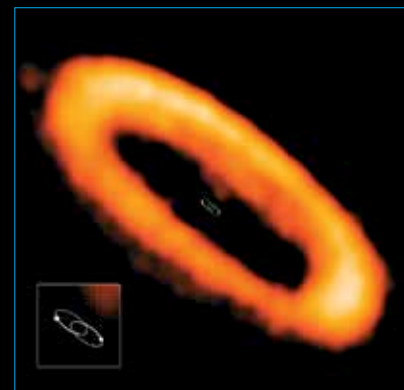
etary discs, and have turned the ALMA (Atacama Large Millimeter/submillimeter Array) instrument to directly study the nearby protoplanetary discs that are suitable for study. The idea is to capture data within the 10 million years of the formation of the system, which is an incredibly short window considering the lifespan of most stars. The instrument can detect the gases directly in the protoplanetary discs, and the researchers use one particular gas, carbon monoxide, as a marker for tracking the dynamics of the accretion discs. By identifying the protoplanets in the accretion discs, and tracking their movements, it is possible to better understand how planets form and find their place within the system. One of the ways scientists are now approaching the problem is to use simulations to track the movements of protoplanetary objects through an accretion disc, and then compare the predicted characteristics to actual observations. Astronomers have found hot Jupiters mostly without other stars around them, but also with, which indicates that the current theories cannot explain all the observations.

Another one of those interstellar oddities are brown dwarfs, objects that straddle the line between being gas giants and stars. Most brown dwarfs are extremely cool and dark, very difficult to spot, but they are bright when they are young. There is some evidence to

suggest that protoplanetary discs form around brown dwarfs as well, and observations from the James Webb Space Telescope can allow researchers to better understand the minimum mass of brown dwarf stars, and identify the tipping point where a planet becomes a star.

Another way in which astronomers can find

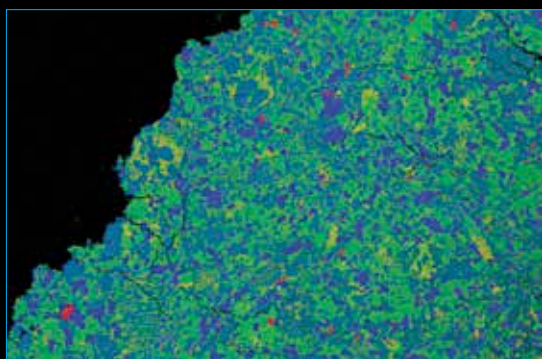
out about the early history of the Solar System is to use the natural time capsules that periodically visit the Sun, comets from the Oort cloud that pour out the matter from the creation of the system every time they become too warm. In 2016, one such comet, Catalina,



A protoplanetary disc around a binary star system

Credit: ALMA (ESO/NAOJ/NRAO), I. Zdzienicka, G. Kennedy, AUI/NSF, S. Dagnello

slingshotted past the Sun, allowing researchers to observe the outgassing. The results from this study have indicated that the outgassing from the comet contains a lot of carbon, which meant in turn that the element was abundant in the outer solar system when the comet was formed. The presence of carbon on the inner solar system planets, such as Earth, and one of the puzzles is exactly how the carbon got from the outer system to the inner system. Researchers now believe that an early perturbation in the orbit of Jupiter, swung comets and other planetary bodies inwards into the Solar System, delivering carbon where it would have otherwise not been available. A series of cometary impacts on a proto-Earth may well enable life as we know it. Further observations of more comets from the outer Solar System is necessary for a better understanding of the processes involved. Researchers are now investigating the exact processes by which these colossal series of impacts actually managed to kick-start life on the planet. **d**



An elemental map of a meteorite. Different colours are different elements.

Credit: Michael Lucas